seconds, while a liquid engine has up to 295 seconds. A cryogenic engine has a specific impulse of more than 460 seconds. A higher specific impulse or energy level meant the need for lesser fuel or more payload mass to orbit. While solid has its uses in the first stage, I argued, to generate the initial thrust, but in a second or upper stage, a liquid engine is more efficient as it has lower acceleration and lower vibration than a solid engine.

When I put forward such scientific facts that none could contest, the group would hear me out, but at the end of it would carry on with simulations of the solid-only configurations as if my arguments were anything but a radio commercial of a toy that came during the break of an adult talk show. When I continued to press, the proponents of solid-only rocket, who constituted a vast majority of the system then, harped on the fact that we did not have a liquid engine, and it would be foolhardy to try out something in PSLV.

At one such meeting, many kept asking, 'Where is the liquid?' I kept insisting that we can make one. Even Kalam did not appear convinced with my arguments. It is another matter that less than six years later, when we successfully ground-tested the Vikas liquid propulsion engine in Vernon in 1985, Kalam, who had moved to the Defence Research and Development Organisation (DRDO), was the first one to congratulate me.

When an overwhelming majority in the PSLV group dismissed the probability of using a liquid engine, the reality was that my team, working with the French in Vernon for more than three years, had learnt the technology. But then, without a project approval, the Vikas engine would not be part of PSLV. To get such an approval, I knew, I had to play another game, which I already did.

It was a Herculean task to procure materials to make the engine, as suppliers to SEP, Vernon, where we learnt to make the engine, would not give them in small numbers. Again, my personal rapport with people like SEP commercial director Rene Morin, and my good old friend Kanwal Grover helped.

After using several contacts and pulling many strings through old contacts, which my team members in Vernon had been diligently